MD-DGFLS: MULTI-DOMAIN DEEPFAKE FORENSIC VERIFICATION CERTIFICATE

IEEE TPAMI / CVPR Technical Forensics Specification v2.2-PROD | Cryptographic Audit Dossier

File Name:	4f55b4fd_synth.png	**Target Domain:**	Face
Audit Timestamp:	2026-09-25 10:26:07 UTC	**SHA-256 Hash:**	b7877434e79a37fbbe0cfdd4...
Forensic Verdict:	**VERIFIED AUTHENTIC MEDIA**	**Confidence:**	**58.4%**
Tamper Type:	Authentic Media Stream	**Bounding Box:**	None

Three-Stage Visual Forensics Pipeline



Stage 1: Source Image



Stage 2: Tamper Overlay



Stage 3: XAI Heatmap

Mathematical Innovations & Multi-Expert Telemetry

CDFAM (Cross-Domain Alignment)	Contrastive Score: 0.4876 (Margin: m=0.3)
ARCC (Entropy Calibration)	Dynamic Calibration Temperature: T = 1.0
TGHFL (Transformer-GAN Cross-Attention)	Cross-Attention Energy: 0.0005
MEFCN (Multi-Expert Consensus)	Face: 71.1% | Medical: 9.5% | Satellite: 6.6% | Regular: 6.4% | Natural: 6.4%
HLRN (Localization Refinement)	Coarse-to-Fine Refinement: 256x256 Native
Frequency Stream (ELA & Laplacian)	ELA Discrepancy: 1.081 | Sensor Noise Var: 1383.1

This certificate is cryptographically anchored and generated by the MD-DGFLS Autonomous Forensics Engine. Verified under IEEE TPAMI standards.